

The Unsustainability of War

Sustainability is often described as meeting the needs of the present without compromising the ability of future generations to meet their own needs. It involves protecting natural resources, preserving biodiversity, and ensuring that economic and social development does not come at the expense of the environment. Discussions about sustainability usually focus on issues such as climate change, pollution, renewable energy, and conservation. However, one major factor that is often overlooked is war. While armed conflicts are mainly seen as humanitarian and political crises, they also have devastating environmental consequences that can last for decades.

Modern warfare causes destruction on a massive scale. Forests are burned, rivers become polluted, wildlife is forced out of its natural habitats, and fertile farmland is left damaged or contaminated. Explosions release harmful chemicals into the air and soil, while military vehicles and equipment consume enormous amounts of fossil fuels, contributing to greenhouse gas emissions. In many cases, the environmental damage continues long after the fighting has ended. Landmines, toxic waste, and destroyed ecosystems can affect both people and wildlife for generations.

The relationship between war and sustainability is clear. A sustainable future depends on healthy ecosystems, clean water, fertile land, and stable climates, yet war directly threatens each of these. The resources required for rebuilding damaged cities and infrastructure also place additional pressure on the environment. As conflicts continue around the world, understanding their environmental impact becomes increasingly important.

When people think about war, they usually imagine destroyed buildings, military vehicles, and human casualties. However, another victim often goes unnoticed: nature. Ecosystems, which have taken hundreds or even thousands of years to develop, can be damaged or destroyed within days of armed conflict. Forests, wetlands, grasslands, rivers, and farmland all suffer as battles take place, leaving lasting consequences for both wildlife and human communities.

One of the most direct ways war damages ecosystems is through bombing and heavy artillery. Explosions destroy vegetation, remove tree cover, and leave large craters in the ground. Entire forests can be burned or cleared to gain military advantage, resulting in the loss of habitats that countless plant and animal species depend on. Once these habitats are destroyed, many species are forced to migrate, while others may not survive at all. Even after the fighting ends, recovering these ecosystems can take decades, and in some cases they may never fully return to their original state.

The ongoing war in Ukraine provides a clear example of this environmental destruction. Large areas of forests, farmland, and protected nature reserves have been damaged by shelling, wildfires, and military activity. Fires caused by explosions have burned thousands of hectares of woodland, destroying habitats and releasing significant amounts of carbon dioxide into the atmosphere. In addition, heavy military vehicles have compacted the soil, making it more difficult for plants to grow and reducing the land's ability to recover naturally.

War also disrupts the delicate balance that keeps ecosystems functioning. Animals often abandon their habitats because of constant explosions, gunfire, and human activity. Birds may stop nesting, mammals migrate to unfamiliar areas, and aquatic species can be affected when rivers and lakes become polluted or blocked by debris. As species disappear from an ecosystem, the relationships between predators, prey, plants, and pollinators are disturbed, creating a chain reaction that affects the entire environment.

A historical example is the Vietnam War, during which large areas of forest were sprayed with Agent Orange, a chemical defoliant used to remove dense jungle cover. Although it achieved military objectives in the short term, it caused severe environmental damage. Millions of hectares of forest were destroyed, biodiversity declined dramatically, and many areas continue to experience ecological and health impacts decades later. This demonstrates that the environmental effects of war often extend far beyond the battlefield itself.

Healthy ecosystems provide essential services such as clean air, clean water, fertile soil, flood protection, and food for both humans and wildlife. When war destroys these natural systems, it weakens nature's ability to support life and recover from future challenges. This is why environmental damage should not be viewed as a secondary consequence of war, but as one of its most serious and long-lasting impacts.

Beyond the visible destruction of landscapes and cities, war leaves behind another serious problem that is often less obvious: pollution. Armed conflicts release large amounts of harmful substances into the air, water, and soil, creating environmental damage that can continue long after the fighting has stopped. Unlike the destruction of buildings, pollution is often invisible at first, but its effects can spread over vast areas and impact both people and wildlife for many years.

Air pollution is one of the most immediate environmental consequences of war. Explosions from bombs, missiles, and artillery produce enormous clouds of dust, smoke, and fine particles that reduce air quality. Military vehicles, aircraft, and ships also consume huge quantities of fuel, releasing carbon dioxide and other pollutants into the atmosphere. In urban areas, damaged factories, power stations, and fuel storage facilities can catch fire, sending toxic gases into the

air. These pollutants not only contribute to climate change but can also cause respiratory illnesses and other health problems for nearby communities.

Water pollution is another major concern during armed conflicts. Bombing can damage water treatment plants, sewage systems, and pipelines, allowing untreated waste and hazardous chemicals to enter rivers, lakes, and groundwater. Oil leaks from damaged storage tanks, destroyed industrial sites, and military equipment can contaminate water supplies that people and animals depend on for survival. Once polluted, these water sources can take years to recover, making access to clean drinking water increasingly difficult.

The 1991 Gulf War is one of the clearest examples of how conflict can cause large-scale environmental pollution. As Iraqi forces retreated from Kuwait, they set hundreds of oil wells on fire and released millions of barrels of oil into the Persian Gulf. Thick black smoke covered the sky for months, reducing air quality across the region, while the oil spill severely damaged marine ecosystems. Fish, seabirds, sea turtles, and other marine life were affected as coastlines became coated in oil. Even decades later, scientists continue to study the long-term environmental consequences of this disaster.

War also contaminates the soil. Explosives, ammunition, and damaged military equipment leave behind heavy metals such as lead, mercury, and copper, along with other toxic substances. These pollutants can remain in the ground for many years, reducing soil fertility and making farmland less productive. Crops grown in contaminated soil may absorb harmful chemicals, creating risks for both human health and food security. In some areas, abandoned military sites and destroyed industrial facilities continue to leak hazardous materials into the surrounding environment long after the conflict has ended.

Another growing environmental challenge is the enormous amount of waste created during war. Collapsed buildings produce millions of tonnes of rubble containing concrete, glass, metal, asbestos, plastics, and other hazardous materials. Disposing of this waste safely requires significant time, money, and resources. If it is not managed properly, harmful particles can spread into the air and nearby water sources, causing further pollution and delaying environmental recovery.

These examples show that pollution is not simply an unavoidable side effect of war—it is one of its most damaging and long-lasting consequences. While cities can eventually be rebuilt, restoring clean air, uncontaminated water, and healthy soil is often far more difficult. The environmental scars left by pollution may remain for decades, affecting ecosystems and human communities long after peace has been restored.

Climate change is one of the greatest challenges facing the world today, and reducing greenhouse gas emissions has become a global priority. However, war works against these

efforts by producing large amounts of carbon emissions and consuming vast quantities of natural resources. Although the environmental costs of conflict are often overlooked, military activities make a significant contribution to climate change before, during, and even after wars.

Modern armed forces rely heavily on vehicles, aircraft, naval ships, and other equipment that require enormous amounts of fuel. Fighter jets, tanks, helicopters, and military transport vehicles burn fossil fuels on a daily basis, releasing large quantities of carbon dioxide into the atmosphere. Unlike many industries that are gradually adopting cleaner technologies, military operations often depend on traditional fuels because of their reliability and efficiency in combat. As a result, prolonged conflicts can produce millions of tonnes of greenhouse gas emissions.

War also increases emissions through the destruction it causes. Bombed buildings, factories, bridges, and roads must eventually be rebuilt, and reconstruction requires huge amounts of materials such as cement, steel, glass, and concrete. Manufacturing these materials is highly energy-intensive and generates significant carbon emissions. In addition, clearing debris, transporting construction materials, and rebuilding infrastructure all require heavy machinery that consumes even more fossil fuels. This means that the environmental impact of war continues long after the fighting has ended.

Another important consequence is that conflict often shifts governments' priorities away from environmental protection. During times of war, countries naturally focus on defence, humanitarian aid, and economic recovery. As a result, climate policies, conservation projects, and investments in renewable energy are frequently delayed or receive less funding. International cooperation on environmental issues can also become more difficult when nations are engaged in conflict, slowing progress toward global sustainability goals.

War therefore creates a cycle that is difficult to break. It damages ecosystems, increases pollution, and accelerates climate change, while at the same time reducing the world's ability to respond to these environmental challenges. From a sustainability perspective, this makes armed conflict not only a humanitarian tragedy but also a major obstacle to creating a healthier and more resilient planet for future generations.

War is often remembered for the lives it claims and the cities it destroys, but its impact on the environment is equally significant and should not be overlooked. As this essay has shown, armed conflict damages ecosystems, pollutes the air, water, and soil, disrupts wildlife, and contributes to climate change. These effects extend far beyond the battlefield, often lasting for decades and making environmental recovery both difficult and expensive.

Sustainability is built on the idea of protecting natural resources so they remain available for future generations. War directly opposes this principle by destroying forests, contaminating rivers, reducing biodiversity, and consuming enormous amounts of energy and materials. The rebuilding process that follows conflict places even greater pressure on the environment, creating a cycle of damage that can continue long after peace has been restored.

Protecting the environment is not only about planting trees or reducing plastic waste; it also requires preventing the large-scale destruction caused by conflict. Governments, international organizations, and individuals all have a role to play in supporting peace, promoting responsible environmental management, and ensuring that ecosystems are protected whenever possible. A truly sustainable future cannot be achieved without recognising the environmental cost of war. By understanding these hidden consequences, societies can make more informed decisions that protect both people and the planet for generations to come.